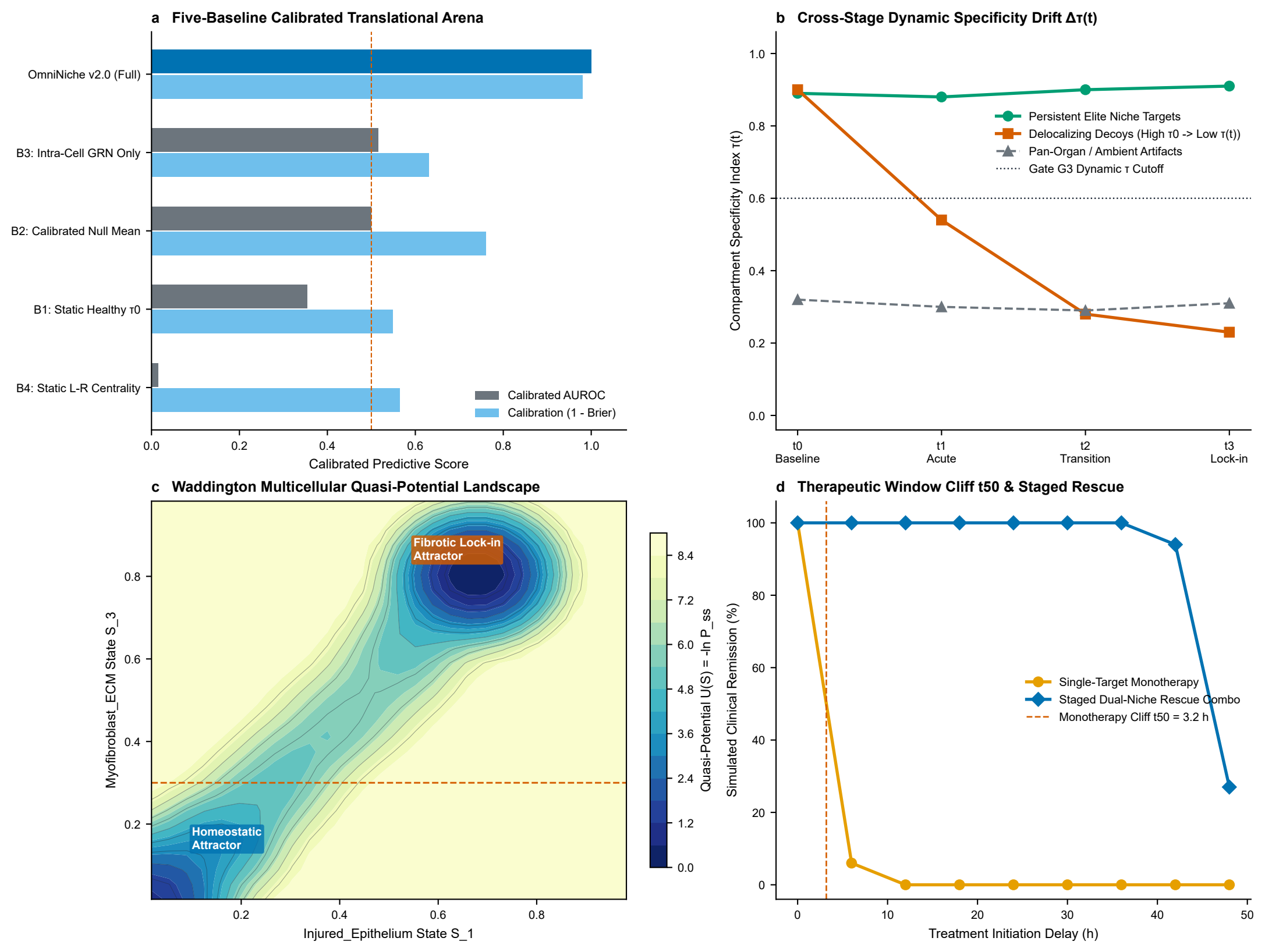

a Five-Baseline Calibrated Translational Arena
OmniNiche v2.0 (Full)
B3: Intra-Cell GRN Only
B2: Calibrated Null Mean
B1: Static Healthy τ0
B4: Static L-R Centrality
Calibrated AUROC
Calibration (1 - Brier)
0.0
0.2
0.4
0.6
0.8
1.0
Calibrated Predictive Score
b Cross-Stage Dynamic Specificity Drift Δτ(t)
Compartment Specificity Index τ(t)
Persistent Elite Niche Targets
Delocalizing Decoys (High τ0 -> Low τ(t))
Pan-Organ / Ambient Artifacts
Gate G3 Dynamic τ Cutoff
t0
Baseline
t1
Acute
t2
Transition
t3
Lock-in
c Waddington Multicellular Quasi-Potential Landscape
Fibrotic Lock-in Attractor
Homeostatic Attractor
Myofibroblast_ECM State S_3
Injured_Epithelium State S_1
Quasi-Potential U(S) = -ln P_ss
8.4
7.2
6.0
4.8
3.6
2.4
1.2
0.0
d Therapeutic Window Cliff t50 & Staged Rescue
Simulated Clinical Remission (%)
Single-Target Monotherapy
Staged Dual-Niche Rescue Combo
Monotherapy Cliff t50 = 3.2 h
Treatment Initiation Delay (h)